

Real Estate Market Analysis & Price Prediction

Data Analysis + Machine Learning



Agenda

- Team Members
- Project Overview
- Dataset Overview
- Data Cleaning & Preprocessing
- Exploratory Data Analysis
- Key Insights
- Machine Learning Model
- Future Enhancements

Team Members

Group-30-Project-02



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Project Overview



Samsung Innovation Campus



Project Overview

This project analyzes real-estate listings to understand market patterns and build machine-learning models that estimate property prices.

The analysis covers two markets:

- Property Sales
- Residential Rentals

Key Tasks

- Prepare and clean the raw housing dataset
- Explore price relationships with size, type, and location
- Identify the strongest drivers of price variation
- Build and evaluate a predictive machine learning model

Tools

- Python (Pandas, numpy) for data analysis
- Scikit-learn for machine learning modeling

Data Source

Kaggle: Egypt Real Estate Data 2026

Dataset Overview



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Dataset Overview

DATASET NAME: propertyfinder.csv

39,713

Listings/ rows

53

OriginalFeatures / columns

2

Markets

Features are indicated by the following rules Groups:



• **Location:** city, district, compound, latitude, longitude



• **Physical size:** area_sqm, bedrooms, bathrooms



• **Price:** price (EGP)



• **Category:** property_type, furnished_status, amenities

Dataset Overview

DATASET NAME: propertyfinder.csv

12

City / Region

New Cairo

Sheikh Zayed

6th of October

New Capital

Maadi

Nasr City

Heliopolis

Zamalek

Al Rehab

Madinaty

Shorouk City

Mostakbal City

Mokattam

Mohandessin

Dokki

Obour City

19

propertie types

APARTMENT



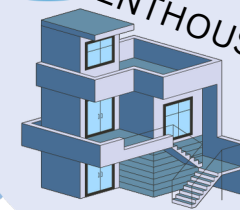
Villa



Cabin



PENTHOUSE





Data Cleaning & Preprocessing



Data Cleaning & Preprocessing

39,713 → 39,585

Rows retained after cleaning

57 → 31

Columns after formatting

0

Duplicated values found

DATA CLEANING WAS IN 3 STEPS:

1

Missing Values

Analyzed missing data across all fields

2

Data Type Cleaning

listed_date → datetime

3

Data Splitting

Splited the data into to data frames (buy, rent)

Data Cleaning & Preprocessing

STEP 1: Handling Missing Values

1

Inspect structure

`df.info()` reviewed to confirm dtypes and non-null counts across all 53 columns.

2

Audit nulls per column

`df.isnull().sum()` run column-by-column, then summed for a dataset-wide total.

3

Result: zero gaps

any column that returned a number of missing values, was set to Unknown until the last step.

4

Imputer Used Before Modeling

A median/most-frequent SimpleImputer routine was used in the last step before training.

Data Cleaning & Preprocessing

STEP 2:

BEFORE
53
original columns



AFTER
31
columns after cleaning

1

Remove Missing-Value Columns

Removed:
video_url
rera
agent_languages

2

Remove Identifiers

Removed :
IDs,
URLs, agent & broker
contact information

3

Remove Irrelevant Fields

Removed:
currency, payment
method,
title, area unit,
reference,
description

4

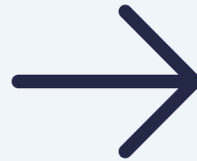
Final Feature Set

53
↓
31
columns Ready for
analysis & machine
learning

Data Cleaning & Preprocessing

STEP 3:

BEFORE
39,585
cleaned listings



AFTER
2 Markets
Buy + Rent

1

buy_df → Sell market

19,802 listings

Residential for
Sale

2

rent_df → Rental market

19,783 listings

Residential for
Rent

EDA

Exploratory Data Analysis



Exploratory Data Analysis

39,585

Property listings analyzed

31

Cleaned features

2

Markets

VIEWING DATA WITH 3 LENSES:

1

**Central Tendency &
Spread**

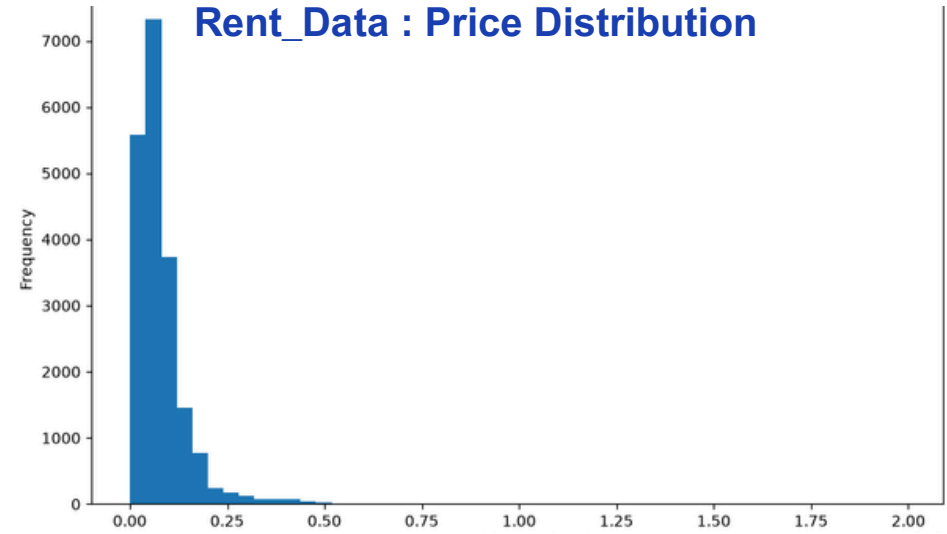
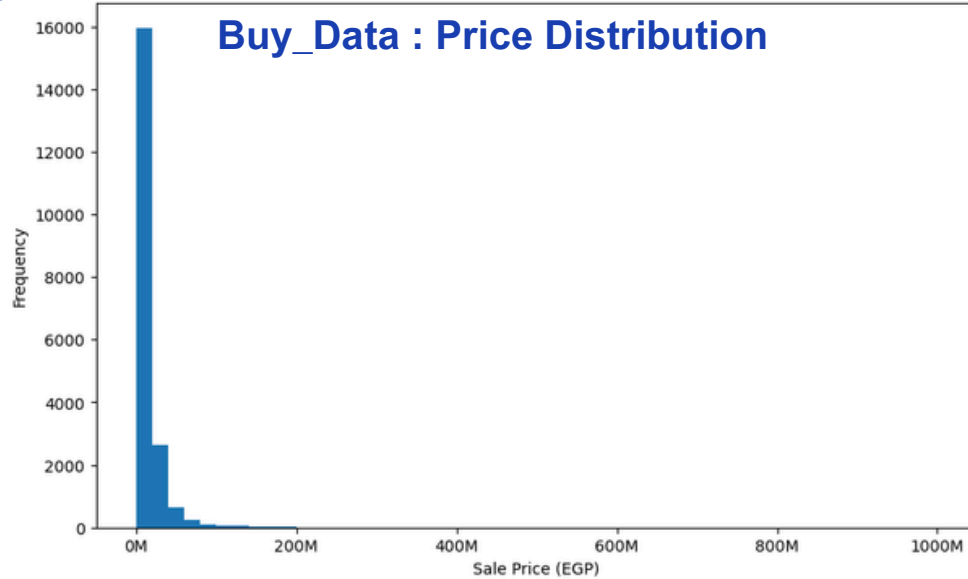
2

insights and patterns

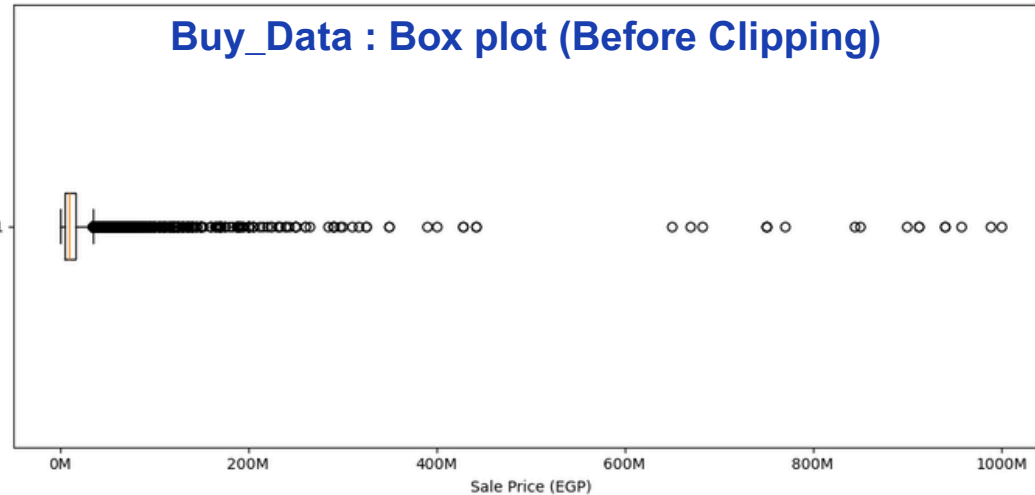
3

Correlation

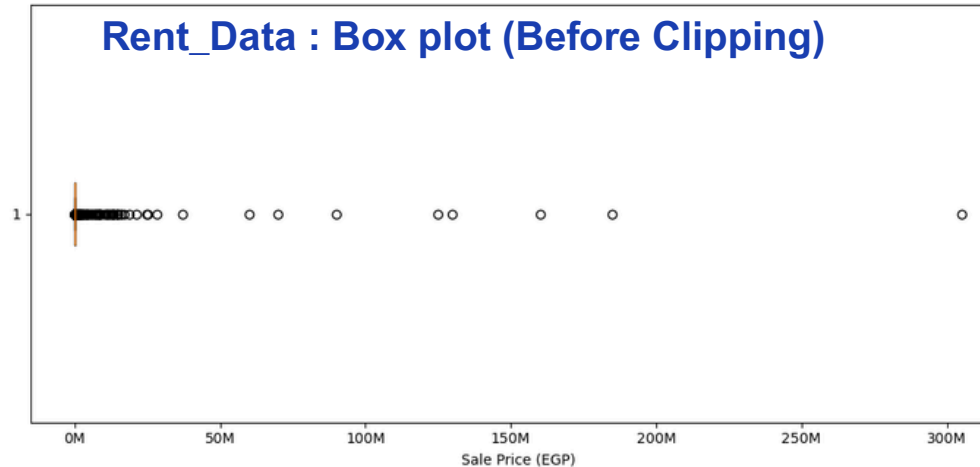
Exploratory Data Analysis



Buy_Data : Box plot (Before Clipping)

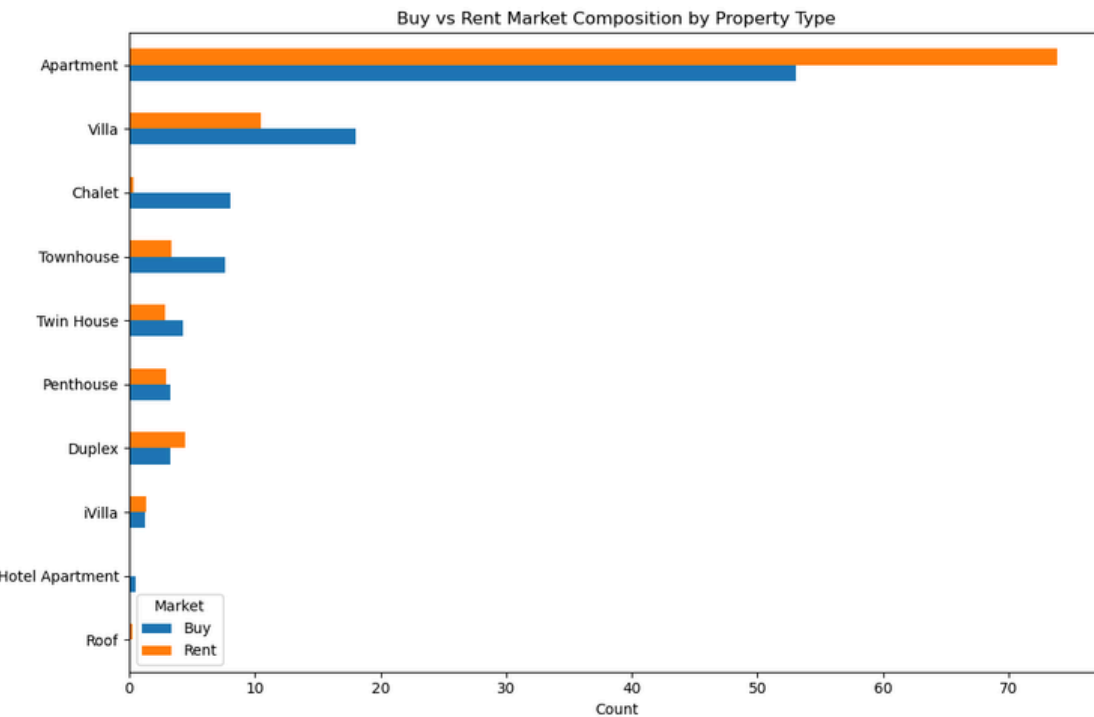


Rent_Data : Box plot (Before Clipping)

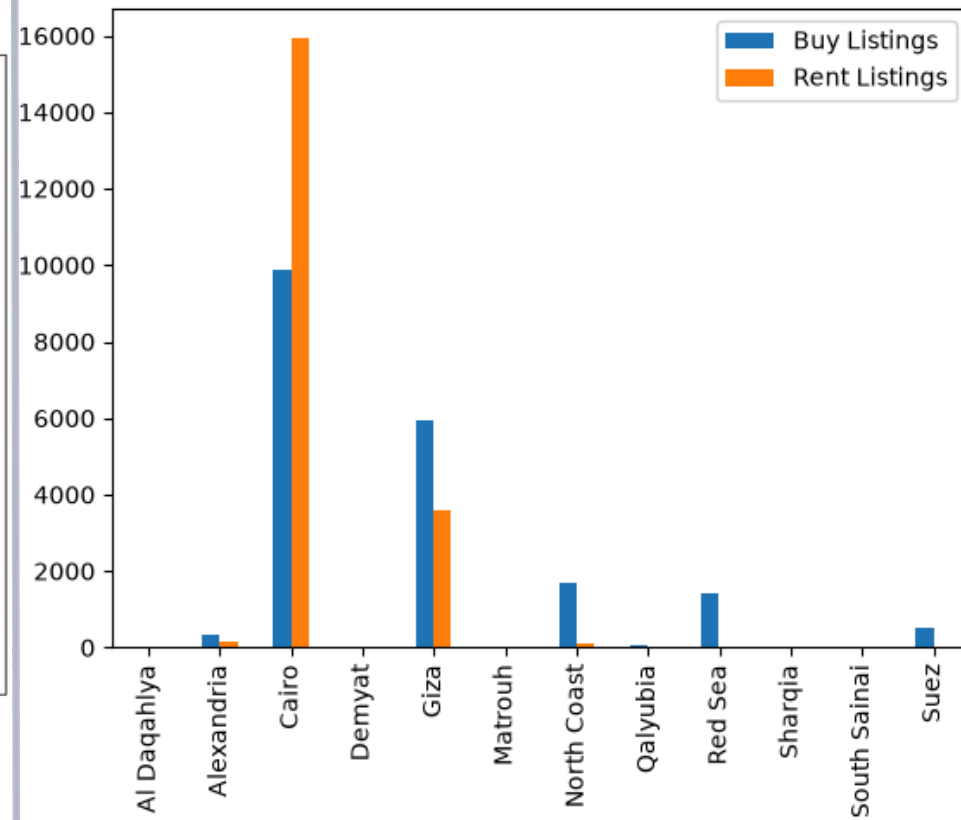


Exploratory Data Analysis

Buy vs Rent : Property Type



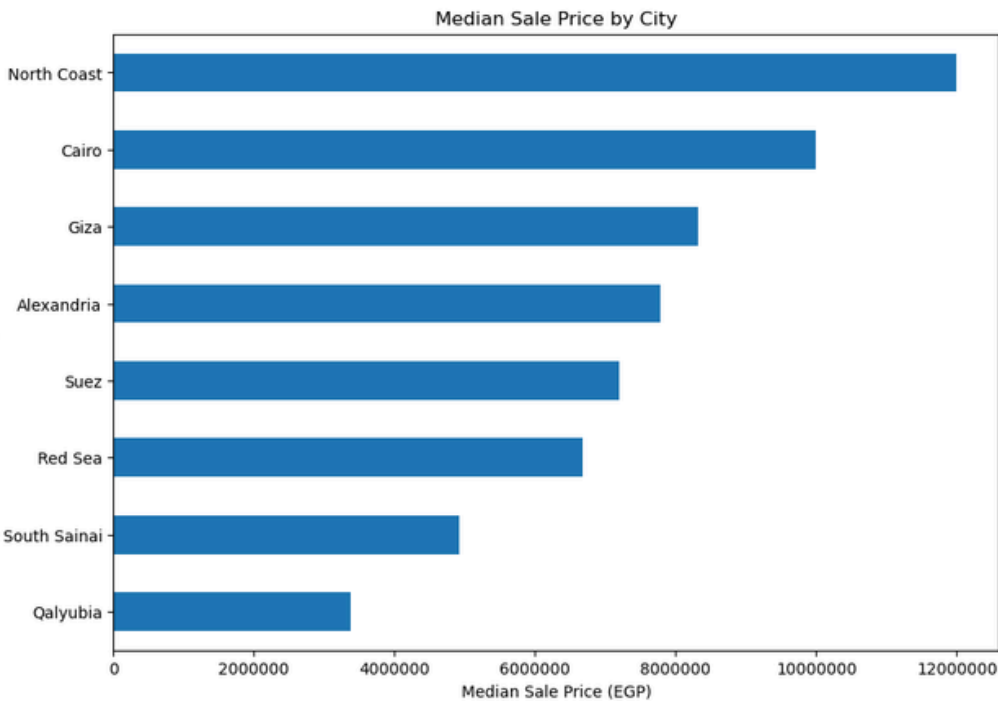
Buy vs Rent : Distribution of Cities



Exploratory Data Analysis

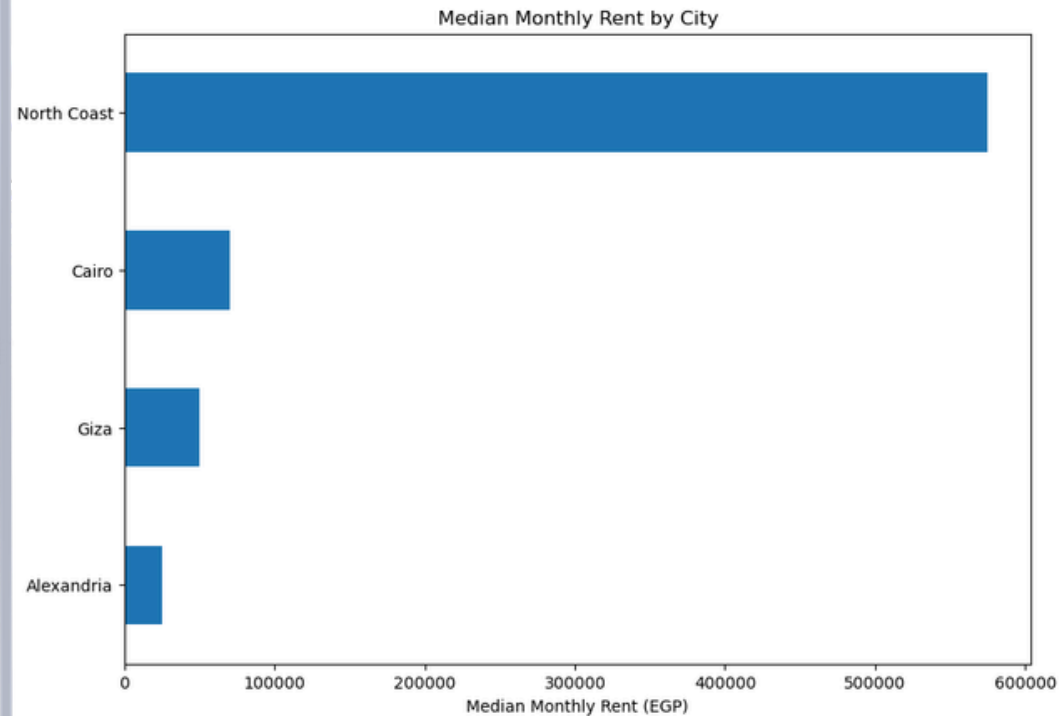
Buy_Data : AVG Price

By City



Rent_Data : AVG Price

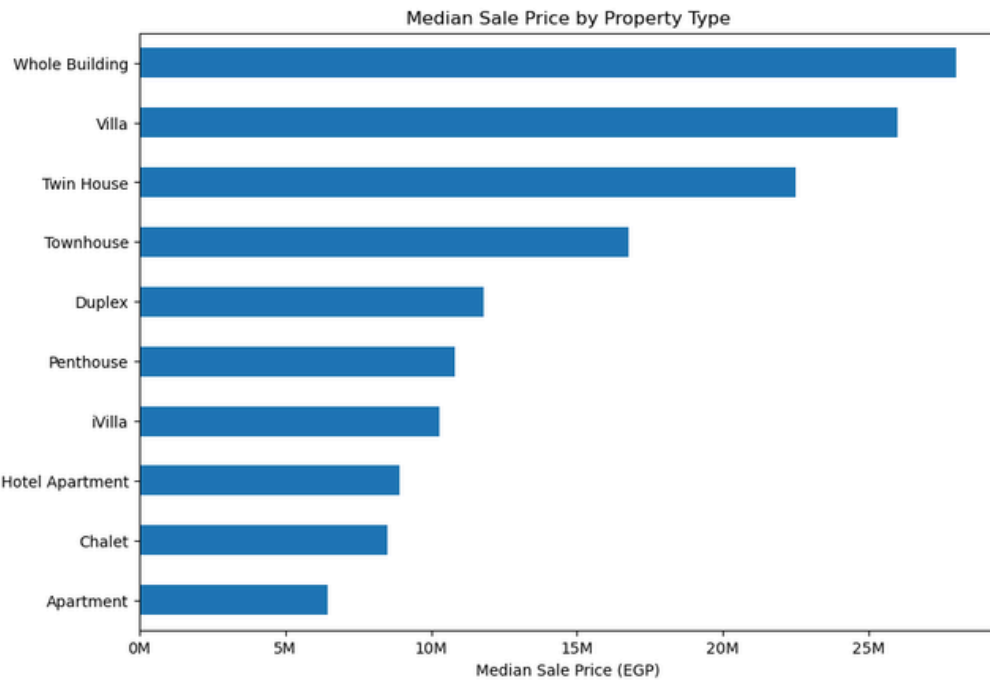
By City



Exploratory Data Analysis

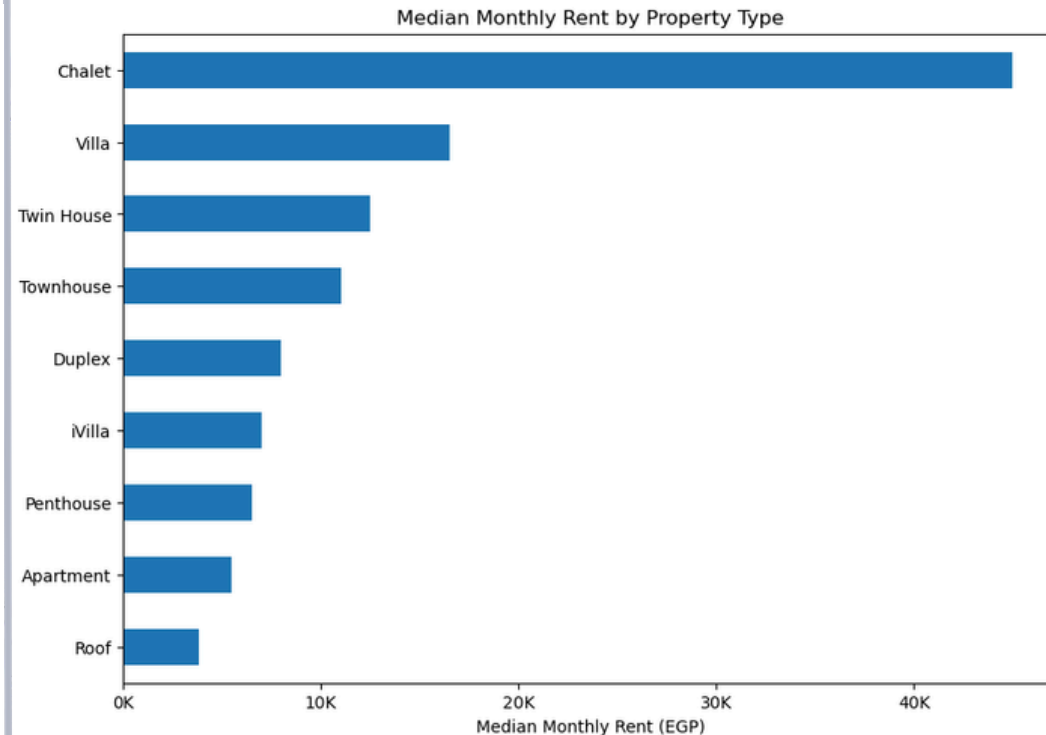
Buy_Data : AVG Price

By Property Type



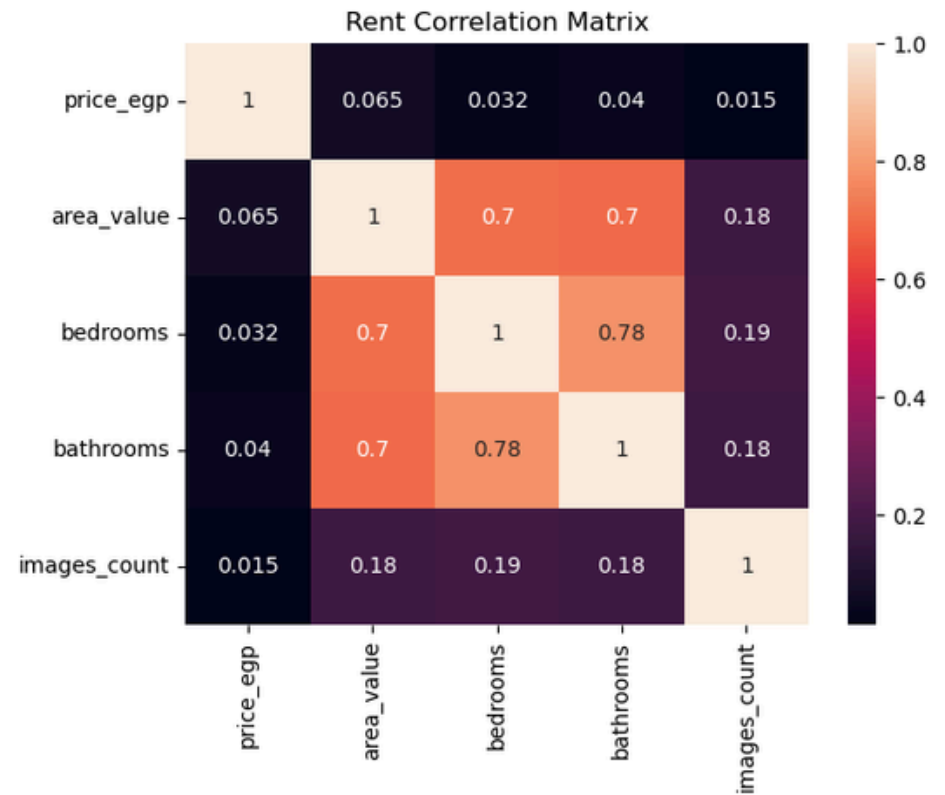
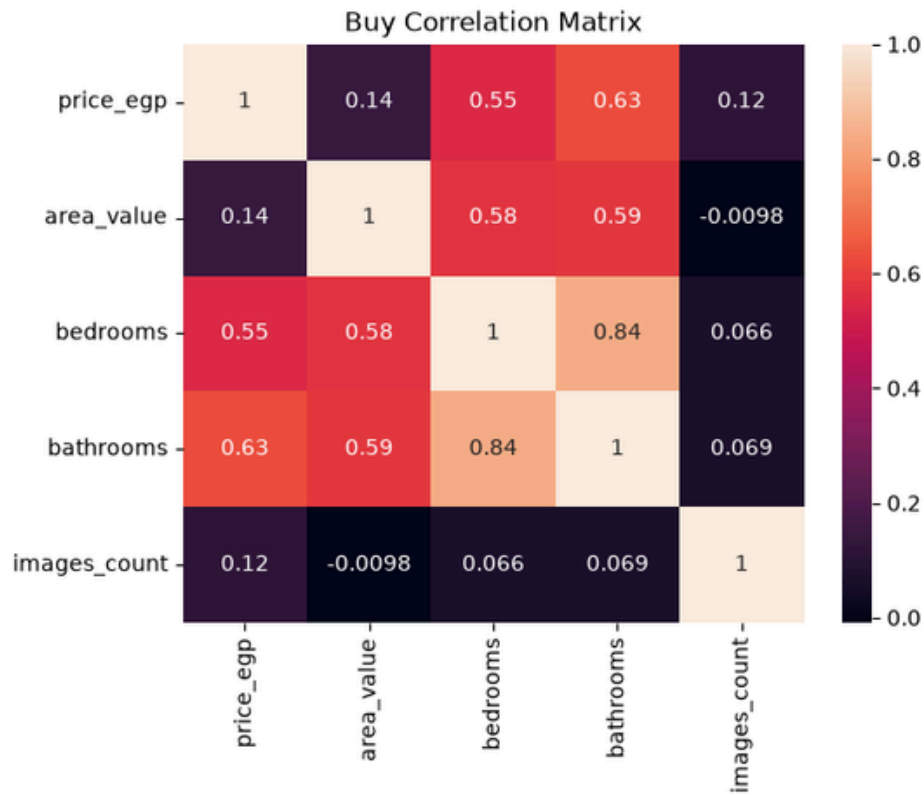
Rent_Data : AVG Price

By Property Type



Exploratory Data Analysis

Data Correlation Matrixes





Key Insights



Exploratory Data Analysis

Extracting insights and patterns

Questions Asked & Patterns Found



Where is the market concentrated?

We found that Cairo dominates the dataset, especially in the rental market, where around 80% of rental listings are located in Cairo.



What affects property prices?

Area, location, and property characteristics are key pricing signals. Their contribution reached around 29.6% of model importance for sale properties and 22.9% for rental properties.



Which property types dominate the market?

Apartments clearly dominate the rental market, representing around 74% of rental listings.



How does pricing behave across the market?

Prices are strongly right-skewed. Most properties are concentrated in the lower and middle price ranges, while a small number of luxury properties have extremely high prices, especially in areas such as the North Coast.



Exploratory Data Analysis

Detecting Skewness & Outliers :

What We Found

PRICE DISTRIBUTIONS WERE RIGHT-SKEWED , And a total of **2760**

Sale Skewness: **1.06**

Sale Outlier : **1604**

Rent Skewness: **6.58**

Rent Outlier : **1156**

Key finding: Rental prices showed much stronger skewness, indicating a long tail of high-value properties.

How We Dealt With It

1. Identified extreme values

- Used statistical analysis and distribution plots to detect unusually high prices and areas.

2. Did NOT blindly remove valid properties

- Extreme prices can represent luxury properties, so they were not automatically treated as errors.

3. Applied log transformation

- Created `log_area` to reduce the influence of highly skewed area values and improve model learning.

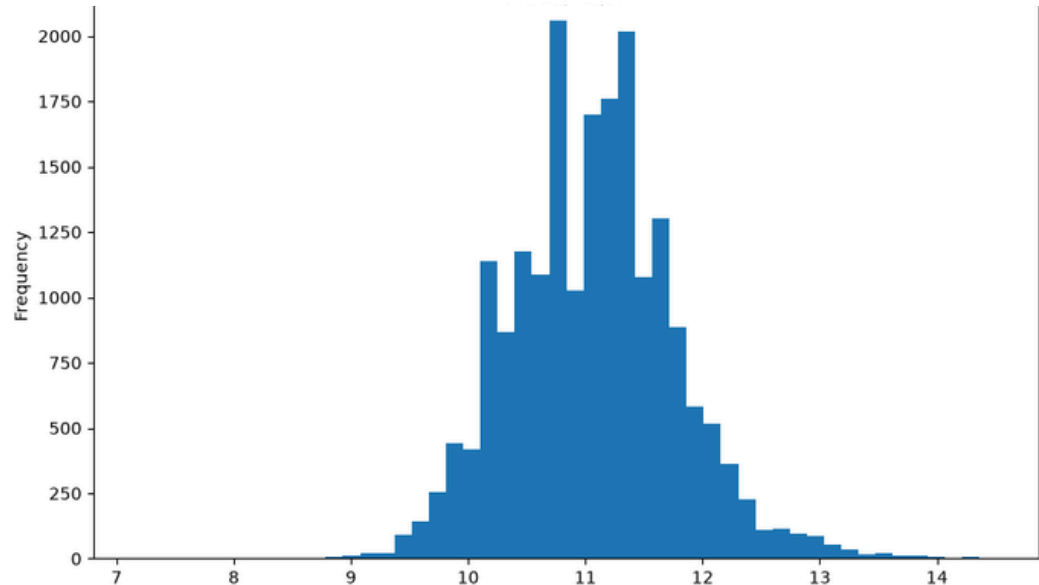
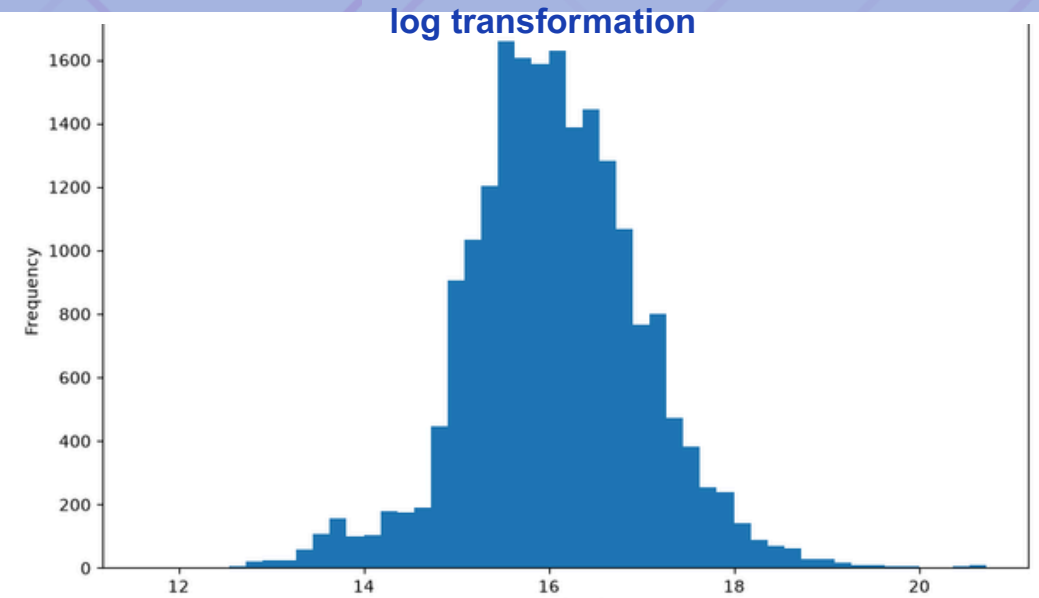
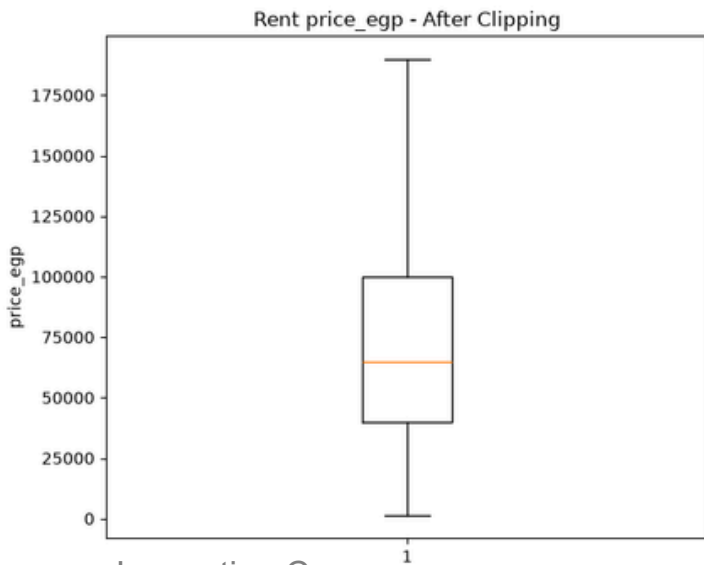
4. Used Random Forest

- The tree-based model is less sensitive to extreme numerical values than many traditional regression approaches.

5. Finally we clipped the Outliers

“So instead of blindly deleting outliers, we handled them carefully to preserve valid luxury properties while improving model stability.”

Exploratory Data Analysis



Machine Learning Model

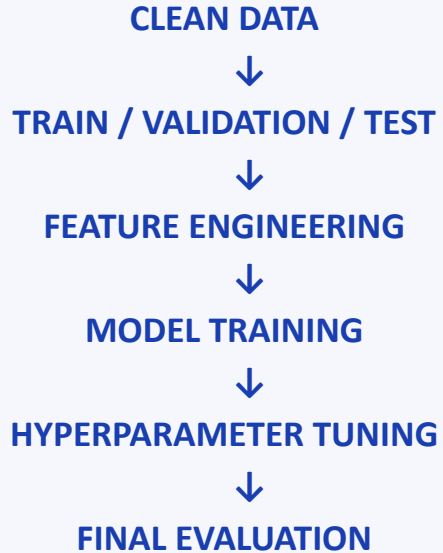


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Model

Modeling Overview :



Problem Type:
REGRESSION

Target: price_egg

Models: Linear
Regression, Random
Forest, Gradient
Boosting ,CatBoost

Model

Modeling Decisions :

- Two separate markets: Sale and Rent
- `'price_per_sqm'` is excluded because it is derived from the target price and would cause target leakage.
- IDs, URLs, contact fields, raw title/description text, `'scraped_at'`, and market-period columns are excluded from predictors.
- We keep a compact structured feature set: **property attributes, city/district, coordinates, amenities count, listing status, and simple date features.**
- Numeric missing values are imputed inside the pipeline using the training data only.
- Categorical variables are imputed and one-hot encoded inside the pipeline.
- A chronological split is used to simulate predicting later listings from earlier listings.

Why Separate Models :

The two markets have different pricing behavior and scales. Separate models allow each market to learn its own relationships.

Model Preprocessing

Feature Engineering :

TOTAL ROOMS :

Bedrooms + Bathrooms

AREA / BEDROOM :

Area $\div$ Bedrooms

BATHROOMS / BEDROOM :

Bathrooms $\div$ Bedrooms

LOG AREA :

$\log(1 + \text{Area})$

LOCATION FEATURES :

Latitude / Longitude

AMENITIES :

Number of Amenities

six new engineered features . and price_per_sqm was NOT used. It is derived from the target price and would cause target leakage.

Hyperparameter Tuning

MODEL TUNING :

Market	Best Configuration
Sale	200 trees, min samples leaf = 2, max features = 0.8
Rent	250 trees, max depth = 20, max features = 0.8

Models were selected using validation R^2 before final testing. Random Forest was selected as the final model for both markets.

Final Model Performance

FINAL MODEL PERFORMANCE :

Sale

Random Forest

R^2 : 0.83 MAE : 2.50M EGP RMSE : 3.98M EGP

Rent

Random Forest

R^2 : 0.84 MAE : 11.24K EGP RMSE : 18.4K EGP

Both models explain more than 83% of the variation in observed prices.

Key Findings & Business Value

KEY FINDINGS :

01 — Market Structure

≈ 50% Sale / 50% Rent

The dataset provides a balanced view of both markets.

02 — Location Matters

49.78% of Sale listings and 80.47% of Rent listings are in Cairo, showing strong geographic concentration.

03 — Property Type

Apartments dominate both markets:

53.07% Sale | 73.90% Rent

04 — Pricing Signals

Area is the strongest model signal, reaching 29.6% importance in Sale and 22.9% in Rent.

KEY FINDINGS :

Price Estimation

Predict expected sale price and monthly rent from property characteristics.

Location-Based Decisions

Identify areas with significantly different price levels and rental demand.

Market Benchmarking

Compare properties against typical market prices instead of relying only on intuition.

Better Investment Decisions

Use data-driven estimates to support property valuation, pricing, and investment screening.



Future Enhancements



Future work

- **More Data :**

Expand the dataset with new listings over time to capture changing market trends and improve generalization.

- **Real-Time Price Prediction :**

Connect the model to live Property Finder data to provide continuously updated price estimates.

- **Advanced Location Features :**

Add distance to schools, universities, transportation, malls, and major landmarks to better capture location value.

- **Improve Outlier Handling :**

Develop dedicated methods for luxury and extreme-value properties instead of treating all high prices the same.

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Education for Future Generations

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